

Multivariate Description

HES 505 Fall 2025: Session 18

Matt Williamson

Objectives

- **Describe** the motivations for dimension reduction with spatial data
- **Distinguish** the differences between PCA and cluster analysis
- **Understand** common metrics for assessing the quality of dimension reduction results
- **Implement** simple PCA and cluster analysis using **R**

What is Dimension Reduction

- Reducing the number of variables without losing important information!!
- Generalizeable (not just for spatial data)

Why Dimension Reduction

- Dealing with (multiple) correlations
- Identifying latent structures in the data
- Parsimony in statistical models

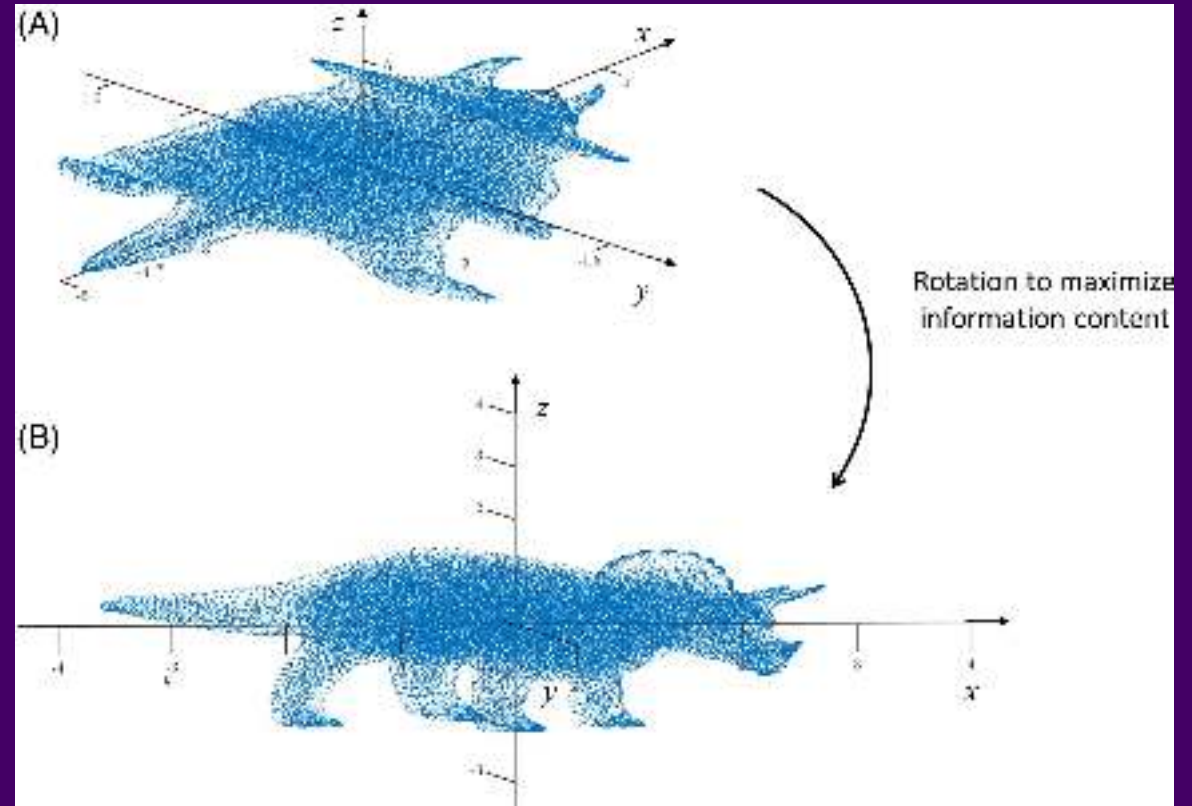
Identifying Gradients with PCA

What is PCA?

- Dimension reduction without info loss
- Components are “new” composed of “parts” of variables
- Variables “load” onto components
- Components are defined to be *orthogonal*

How does PCA work?

- Creates new “coordinate systems” that maximizes variance on that axis
- “Rotation matrix” maps original variables onto new axes



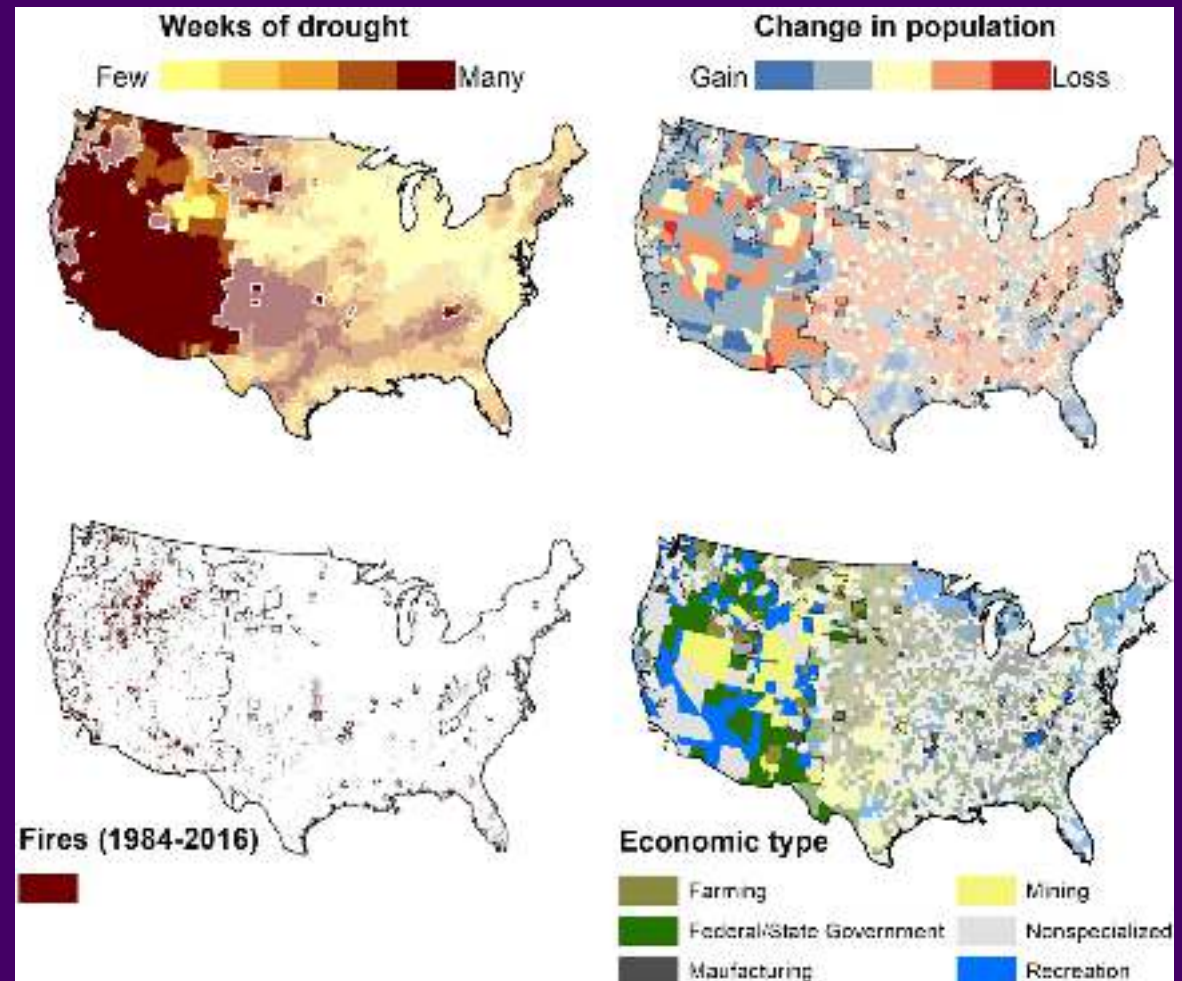
How can we tell how well PCA worked?

- Variance Explained per component
- Cumulative variance explained (scree plots)

Unsupervised Classification with Cluster Analysis

Unsupervised Classification

- No “outcome” variable
- Consistent “patterns” in the data = clusters
- Reduces (a lot of) information into categorical classes

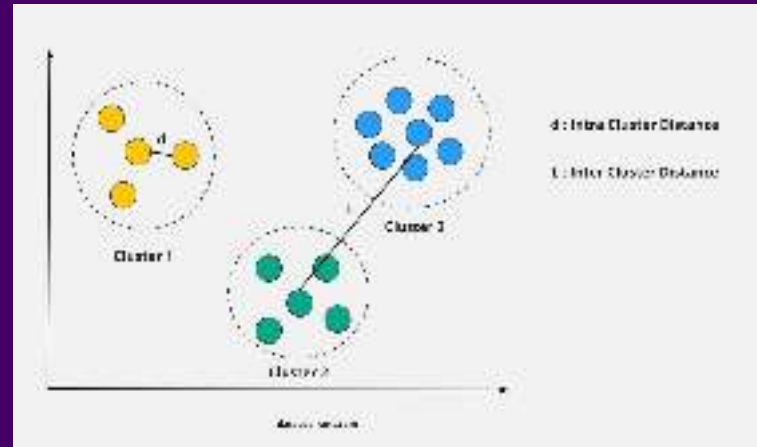


How Does Cluster Analysis work?

- Data in multivariate space
- Cluster centers assigned to minimize distance to observations and maximize distance between centers
- Variation in how centroids are assigned and stopping rules

Evaluating the Quality of Our Clusters

- Explained inertia - *within* cluster variation (0:1, lower = better)
- Silhouette index - overall structure of the data (-1:1, higher = better)



Final Thoughts

Things to Consider

- Spatial autocorrelation
- Newer Algorithms
- What is your Goal?

